Modern Academy
for Engineering and Technology
Basic Science Department

Academic Year: 2016-2017
Semester: Fall
Exam Date: 15/1/2017

QUESTIONS FOR THE FINAL WRITTEN EXAM

Subject: (MTH203) Differential Equations and Transforms **Spec:** Comp. and Comm.
Examiners: Dr. Ashraf Taha and Prof. Dr. Aly Essawi **Time:** 3 hours

Number of Pages: 2 Number of Questions: 5 Attempt all questions

Question 1 (14 points)

Solve the following differential equations:

a) $\left(1+e^{x}\right)\dfrac{dy}{dx}=ye^{x}.$ (2 points)

b) $\left(x+\cos y\right)dx+\left(\tan y-x\sin y\right)dy=0.$ (4 points)

c) $x^{3}\dfrac{dy}{dx}=y^{3}+xy^{2}-x^{2}y.$ (4 points)

d) $\dfrac{dy}{dx}-\dfrac{x+1}{2x}y=-\dfrac{3}{x}y^{3}.$ (4 points)

Question 2 (16 points)

Find the general solution of the following differential equations:

a) $\dfrac{d^{3}y}{dx^{3}}-2\dfrac{d^{2}y}{dx^{2}}-5\dfrac{dy}{dx}+6y=0.$ (4 points)

b) $\dfrac{d^{2}y}{dx^{2}}-\dfrac{dy}{dx}-2y=6x+6e^{-x}.$ (4 points)

c) $\dfrac{d^{2}y}{dx^{2}}+y=\tan\left(x\right)\sec\left(x\right).$ (4 points)

d) $x^{2}\dfrac{d^{2}y}{dx^{2}}+x\dfrac{dy}{dx}-y=x,$ given that $y=x$ is a homogenous solution. (4 points)

Question 3 (16 points)

a) ***Solve the following integration using Laplace transform:***

$\int_{0}^{\infty}e^{-3t}\ \sin^{2}t\ \ dt.$ (3 points)

b) ***Solve the following differential equation using Laplace transform:***

$\dfrac{d^{2}y}{dx^{2}}-6\dfrac{dy}{dx}+8y=e^{3t};\qquad y(0)=0,y'(0)=2.$ (4 points)

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c) ***Find the Laplace transform of the following functions:***

i. $f(t)=\int_0^t \tau \sin\tau \; d\tau.$ (3 points)

ii. $f(t)=\sin 5t\cos 3t+\cosh 3t.$ (3 points)

iii. $f(t)=u_4(t)\, e^t\,.$ (3 points)

Question 4 (16 points)

a) ***Find the inverse Laplace transform of the following functions:***

i. $F(s)=\ln\left(\dfrac{s^2+16}{(s-2)(s-3)}\right).$ (4 points)

ii. $F(s)=\dfrac{s}{s^2+4s+8}.$ (4 points)

iii. $F(s)=e^{-2s}\left(\dfrac{1}{s^2-10s+16}\right).$ (4 points)

b) ***Represent the following function*** $f(x)=x^2,\ 0\le x\le 2$ ***as Fourier Sine Series*** (4 points)

Question 5 (8 points)

a) ***Prove the recurrence relations*** $P_n(x)+2x\,P_n'(x)=P_{n+1}'(x)+P_{n-1}'(x).$ (4 points)

b) ***Evaluate the integral*** $\int_{-1}^{1} x\, P_1(x)\; dx.$ (2 points)

c) ***Evaluate the integral*** $\int_{-1}^{1} P_2^2(x)\; dx.$ (2 points)

GOOD LUCK